NIR-to-RGB-C: A Corruption Robustness Benchmark for NIR-to-RGB Spectral Translation

S. No.	Dataset	Link	Details	Dataset Exploration Sample image Details
1	IDD-AW	https://idd.insaan.iiit.ac.in/dataset/download/	Dataset designed for autonomous driving in unstructured traffic	5,000 high-resolution RGB-NIR image pairs annotated for rain, fog, low light, and snow.
2	OMSIV/SSMID	https://drive.google.com/file/d/1KDi-JJLgMeM6iVN6WXbsAxSzL-_6lIJv/view	Dataset containing images captured using a single-sensor multispectral RGB-NIR (Red, Green, Blue, Near-Infrared) camera	Contains outdoor scenes with heavy vegetation captured under various sunlight conditions and urban environments Count NIR images.-**500** Count RGB images.--**500**
2	RANUS	https://drive.google.com/file/d/1WJ7rcCeMBPy9Qb2c_pwI0rUIlFqotkzT/view	RGB and NIR urban scene dataset (RANUS) for vehicle applications, consisting of 40k precisely aligned scenes and 4k manually annotated masks	40,000 urban road scene images Spatially aligned **RGB and NIR image pairs** Captured from urban driving environments Used to train the chromaticity estimation network No dense semantic annotations required for all 40k images

INCLUDE THE RESULTING IMAGES, DATASET IMAGES

Dataset Table

Dataset	Modality	Size / pairs	Resolution	Split	Ground truth	Purpose	Priority	Notes
SSMID / OMSIV	RGB-NIR single-sensor restoration	Updated OMSIV has 500 registered RGB/RGB-NIR images; restorations split 400 train / 100 test	Registered: 320×580; raw/h5 larger	400/100 official	RGB target	**Primary/secondary**	Required if VCIP works; fallback if VCIP fails	Official SSMID page provides OMSIV details and folders (Xavysp)
EPFL RGB-NIR Scene Dataset	RGB + NIR scene pairs	477 images in 9 categories	TIFF 1024×768	No official train/test; use 70/10/20 or 333/144 from prior papers	RGB	Secondary	Recommended	Good R filters and registered with SIFT/RA cross-dataset validation; captured with visible/NI NSAC (EPFL)

RGB2 NIR / p2irc	RGB→ NIR aerial crop transl ation	[NEEDS VERIFICAT ION] for packaged pairs	Depends on Micasense processing	Not fixed	NIR target, not RGB target	Optional	Only for extensio n	Official repo provides registrati on and pix2pix-style translatio n code, not a clean NIR-to-RGB-C-ready dataset (GitHub)
Real-NAID	NIR-assist ed denois ing	Real-world dataset	[NEEDS VERIFICA TION]	Provided by authors	Noisy/ clean denoisi ng setup	Optional evidence	Not primary	Useful to justify real NIR/RGB noise, not direct NIR-to-RGB translatio n (GitHub)
deep NIR	NIR+R GB / synthe tic NIR / fruit detecti on	Project page says reprocesse d nirscene/S EN12MS plus sweet pepper, with 127k oversample d pairs from nirscene	Mixed	Not NIR-to-RGB standard split	NIR/R GB + boxes	Optional	Avoid in first month	Better for RGB→NI R or detectio n, not direct NIR-to-RGB-C first version (Inkyu)

LLVIP	Visible + thermal infrared	15,488 aligned pairs	Varies	Provided	Visible + thermal, pedestrian labels	Optional downstream	Avoid as core	Thermal 8–14 μm, not NIR; good for later multimodal robustness, not NIR-to-RGB colorization (BUPT AI CZ)
RASMD / RAMSD	RGB + SWIR driving	Pixel-wise registered SWIR/RGB			RGB/SWIR	Strong future extension	Not first version	SWIR driving adverse-weather dataset; useful for CVPR-strength extension, not first 1-month version (STL Lab)

First Start with 2 Dataset - SSMID, EPFL

Task 1:

EPFL

1. Count NIR images.-**477**

2. Count RGB images.--**477**

3. Match pairs by filename stem or provided list.-**477 successfully matched pairs (0 unpaired)**

4. Detect unreadable files.-**0**

5. Check channel count.-**4-channel (RGB+NIR)**

6. Check resolution distribution.-**All 477 pairs share a resolution of 139×94.**

7. Compute pixel statistics.RGB: **B=62.2, G=65.0, R=64.6**

NIR: B=62.0, G=64.7, R=64.3

8. Compute Sobel/Canny edge maps for NIR and RGB-luminance.**Computed successfully for all 477 pairs.**

9. Estimate edge correlation.**Sobel: mean=0.377 (min=0.124, max=0.675)**

Canny: mean=0.214 (min=0.067, max=0.389)

10. Save 20 sample pair visualizations.-**Computed successfully**

11. Classes-**9 classes**

- **country (52 pairs)**
- **field (51 pairs)**
- **forest (53 pairs)**
- **indoor (56 pairs)**
- **mountain (55 pairs)**
- **oldbuilding (51 pairs)**
- **street (50 pairs)**
- **urban (58 pairs)**
- **water (51 pairs)**

OMSIV DATSET

Count NIR images.-**500**

Count RGB images.--**500**

3. Match pairs by filename stem or provided list.-**100 successfully matched pairs**

4. Detect unreadable files.-**0**

5. Check channel count.-**3-channel (RGB)**

1-channel (NIR)

6. Check resolution distribution.-**All 500 pairs share a resolution of 580×320**

7. Compute pixel statistics.RGB:

```
NIR (n=500):
  mean_of_means: 117.8632
  std_of_means: 24.3924
  mean_of_stds: 60.7803
  global_min: 0.0000
  global_max: 250.0000
  samples: 500.0000
RGB: 100%|

RGB (n=500):
  mean_of_means: 112.8824
  std_of_means: 28.2275
  mean_of_stds: 67.1008
  global_min: 0.0000
  global_max: 255.0000
  samples: 500.0000
```

8. Compute Sobel/Canny edge maps for NIR and RGB-luminance.**Computed successfully**

9. Estimate edge correlation

Sobel Correlation (n = 477)

- Mean = **0.377**
- Min = **0.124**
- Max = **0.675**

Canny Correlation (n = 477)

- Mean = **0.214**
- Min = **0.067**
- Max = **0.389**

10. Save 20 sample pair visualizations.-**Computed successfully**

11.Category- Outdoor Multispectral Images with Vegetation

Task 2: For corruption – corrupt only the **NIR input** and keep the **RGB target clean**

10 corruptions × 3 severities

Corruption	Category	Physical motivation	Implementation idea	Severity	Library
Gaussian noise	Noise	Electronic sensor noise	x + N(0, σ)	3	NumPy/OpenCV

Poisson/photon noise	Noise	Low photon count	np.random.poisson(x * scale) / scale	3	NumPy
Shot noise	Noise	Signal-dependent noise	variance proportional to intensity	3	NumPy
Impulse noise	Noise	Dead/hot pixels	salt-pepper mask	3	scikit-image/custom
Motion blur	Blur	Moving platform/camera	linear kernel convolution	3	OpenCV
Defocus blur	Blur	Lens/focus error	disk/gaussian kernel	3	OpenCV
Fog/haze	Atmospheric	scattering / contrast loss	atmospheric blend	3	Albumentations/custom
Rain streaks	Weather	outdoor NIR camera conditions	streak overlay + blur	3	Albumentations/custom
JPEG compression	Digital	transmission/storage loss	re-encode JPEG quality	3	PIL/OpenCV
Bit-depth reduction	Sensor/digital	quantization loss	reduce to 6/4/3 bits	3	NumPy

—----------Computed successfully

Task 3: Baseline Models that you have to run: make sure to use same weight/parameter

Model	Code	Pretrained?	Input	Compatibility	Difficulty	Priority	Risk

NIR repeat-to-RGB	Your code	Not needed	1-channel NIR	All datasets	Easy	Must-run	None
Histogram matching	Your code	Not needed	NIR + train RGB stats	All datasets	Easy	Must-run	Weak baseline
Pix2Pix	Official / PyTorch	Usually train yourself	Paired NIR/RGB	All paired datasets	Easy-medium	Must-run	Need paired formatting (GitHub)
U-Net L1/L2	Your code	Train yourself	Paired NIR/RGB	All paired datasets	Easy	Must-run	May underperform
color_restorer	Official OMSIV repo	Check repo	RGB-NIR restoration input	OMSIV mostly	Medium	Must-run for OMSIV	Dataset-specific; TF/Keras issues (GitHub)
MCFNet	Official repo	Check repo	NIR→RGB	VCIP likely	Medium-high	Must-run if trainable	Pretrained availability risk (GitHub)
ColorMamba	Official repo	Google Drive pretrained listed	NIR→RGB	VCIP	Medium-high	Must-run	Dependency risk (GitHub)
SF-GPT	Official repo		NIR→RGB	VCIP/SSMID examples	Medium-high	Must-run	New repo; check inference scripts (GitHub)

HAQAGen	Project page + code link	Can get code from sir	NIR→RGB	VCIP/OMSIV/RGB2NIR	High	Optional strong	Recent; may require adaptation (Rajeev Ranjan Dwivedi - AI Researcher)
RING	Official repo	Check checkpoints folder	NIR→gray→RGB	ICVL/TokyoTech/IDH by default	High	Optional	Dataset mismatch with VCIP/OMSIV (GitHub)
CoColor	Paper verified	Official code not verified here	NIR→RGB		High	Literature only unless code found	Code risk (arXiv)
MPFNet	Paper verified	Official code not verified here	NIR→RGB		High	Literature only unless code found	Code risk (arXiv)
AAFSTNet	Not reliably verified	[NEEDS VERIFICATION]	NIR→RGB		High	Do not depend on it	Name/code ambiguity

Task 4:Main Benchmark Methods-

1. **Download OMSIV and EPFL.**
2. **Audit images and create clean manifests.**
3. **Define fixed clean test split.**
4. **Generate corrupted NIR inputs only.**
5. **Keep RGB ground truth unchanged.**
6. **Run all models on clean and corrupted NIR.**
7. **Compute clean metrics.**

8. Compute corrupted metrics.
9. Compute metric drop and robustness ranking.
10. Save qualitative failure examples.
11. Release benchmark scripts and corruption configs.

Must remember these things that we will show : Robustness definitions

Term	Definition
Clean performance	Metric on clean NIR input and clean RGB target
Corrupted performance	Metric on corrupted NIR input and same clean RGB target
Absolute drop	CleanMetric - CorruptedMetric for PSNR/SSIM; CorruptedLPIPS - CleanLPIPS for LPIPS
Relative drop	100 × (Clean - Corrupted) / Clean
Per-corruption robustness	Average over severities for one corruption
Per-severity robustness	Average over corruptions at one severity
Worst-case score	Worst metric over all corruptions/severities
AUC severity score	Area under metric-vs-severity curve
mCE-style score	Normalize each model's corruption error by a reference baseline such as Pix2Pix

The Metrics that we have to show (Must be there)-

Metric	Why
PSNR ↑	Pixel fidelity
SSIM ↑	Structure preservation
LPIPS ↓	Perceptual similarity
ΔPSNR / ΔSSIM / ΔLPIPS	Robustness degradation
Mean corrupted PSNR/SSIM/LPIPS	Overall corruption performance
Worst-corruption score	Safety/reliability signal
FPS / inference time	Practical deployment
Params	Efficiency comparison

FID ↓	Distribution realism; use carefully because small test sets can make it unstable
Delta-E ↓	Color difference; useful for colorization
Color histogram distance ↓	Hue/statistics drift
Edge preservation ↑	Whether texture/structure survives
FLOPs / GPU memory	Efficiency table
AUC severity	Clean severity-curve summary

- Analogical Image Translation for Fog Generation
- Deflicker-Awared Foggy Video Generation for UAV Defogging

- SynFog: A Photo-realistic Synthetic Fog Dataset based on End-to-end Imaging Simulation for Advancing Real-World Defogging in Autonomous Driving
- A Systematic Review on Foggy Datasets: Applications and Challenges

NIR-REPEAT

Metric	Value
Clean PSNR	24.1248
Clean SSIM	0.7481
Clean LPIPS	0.4350
Mean Corrupted PSNR	19.0397
Mean Corrupted SSIM	0.5676
Mean Corrupted LPIPS	0.6158
Delta PSNR	5.0851
Delta SSIM	0.1805
Delta LPIPS	0.1809
FPS (CPU)	1503.53

PIX2PIX

Clean images

Metric	**Value**
PSNR	**47.1126**

SSIM	**0.9979**
LPIPS	**0.000610**
Delta E	**0.2605**
Color Histogram Distance	**1.3927**
Edge Preservation	**0.9994**
Number of Images	**49**

Weights and parameters:

Generator ArchitectureU-Net (8 downsampling layers, 256x256)

Discriminator Architecture70x70 PatchGAN

Generator Filters :(ngf)64

Dropout Rate0.5

Output activation-tanh

Discriminator Filters (ndf)64

Initial Learning Rate0.0002

Optimizer-Adam - beta(0.5, 0.999)

Learning Rate-100 epochs constant, 100 epochs linear decay

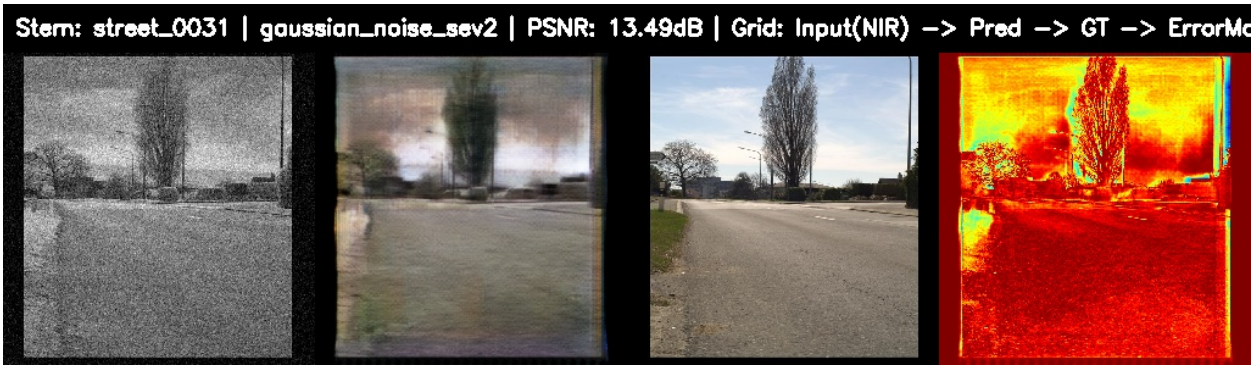
Loss:GAN Loss

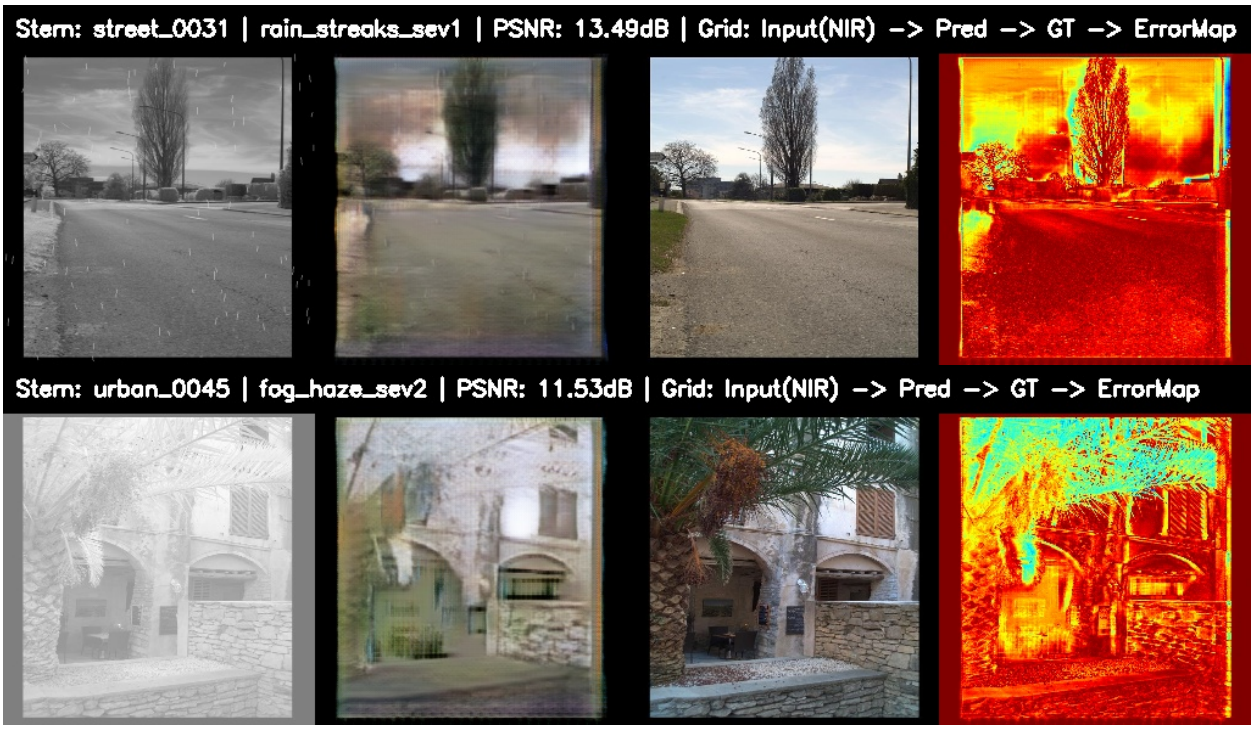
Vanilla GAN (BCE)

L1 Loss Weight : 100.0



Failure:





Followed this https://github.com/vision4robotics/DFVG after going through the paper Deflicker-Aware Foggy Video Generation for UAV Defogging

Fog generation-Ground truth





used Atmospheric Scattering

$$t(x) = e^{-\beta d(x)}$$

Beta-2.5

The standard atmospheric model is:

$$I_{fog}(x) = I_{clear}(x)t(x) + A(1 - t(x))$$

where:

- A = atmospheric light
- I_{clear} = original image
- I_{fog} = foggy image

This is applied pixel-by-pixel

Model-Colour restorer (CDNET)

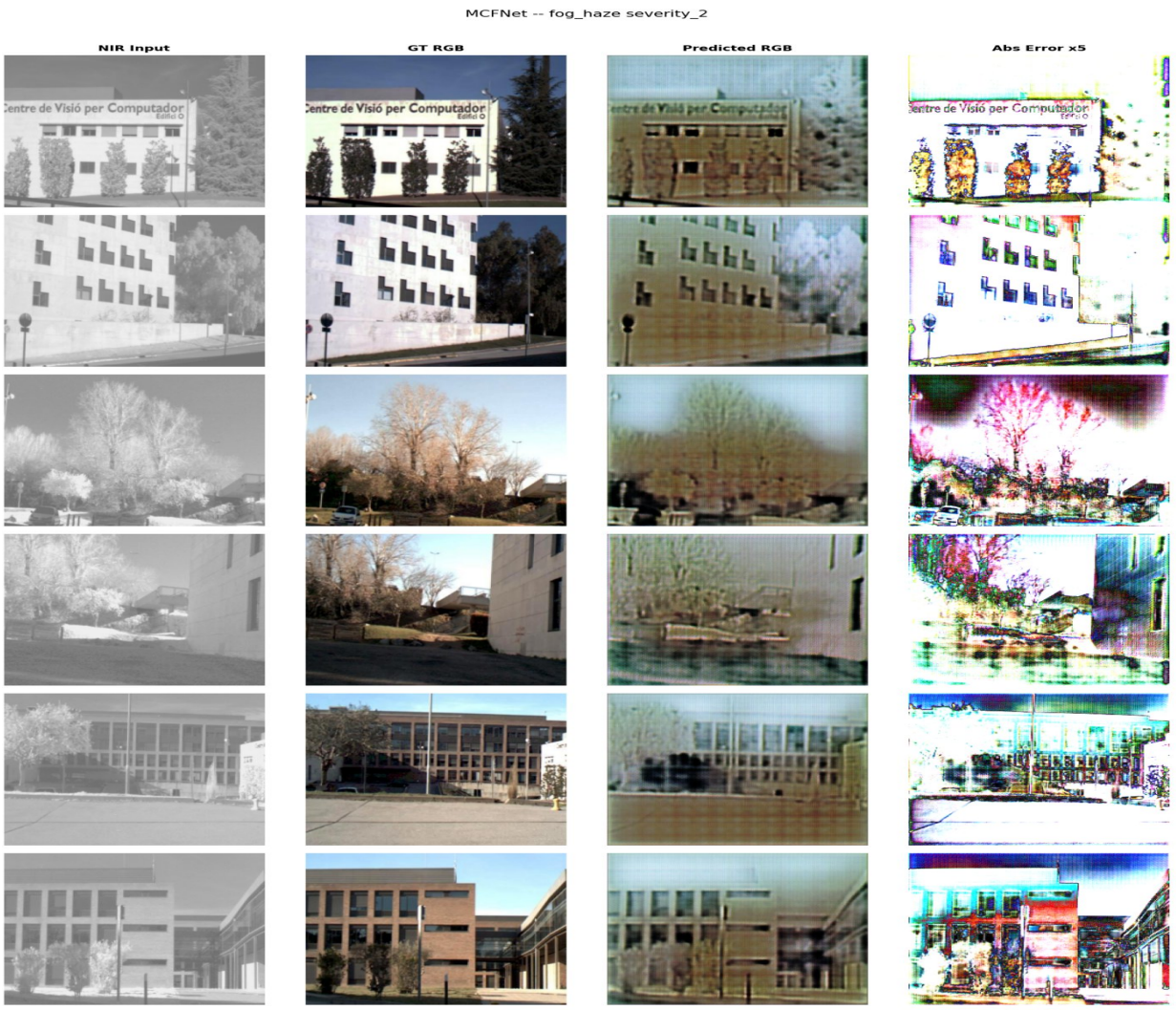
Metrics	clean	corrupted	change
PSNR	16.53	15.53	-0.99
SSIM	0.6412	0.5825	-0.0588
FID	111.42	148.56	37.12
DELTA E	32.16	37.37	5.21
EDGE PRESERVATION	0.37	0.310	-0.06
FPS	117.73	115.26	-2.47



MCFNet

Metric	Clean Value	Corrupted Global Value	Difference (Δ)

PSNR	**10.9549**	**11.1897**	**+0.2348**
SSIM	**0.3490**	**0.3228**	**-0.0262**
Delta E	**62.8718**	**61.8089**	**-1.0629**
Histogram Distance	**0.4213**	**0.4277**	**+0.0064**
Edge Preservation	**0.1374**	**0.1044**	**-0.0330**
Inference Time (ms)	**11.7928**	**9.4535**	**-2.3393**
FPS	**84.7976**	**105.7810**	**+20.9834**



SF-GPT(SSMID dataset)

	CLEAN	CORRUPTED
PSNR	18.5888	19.6067
SSIM	0.6909	0.6786
DELTA E	11.5895	10.1141
EDGE PRESERVATION	0.8338	0.8317
FPS	6.5410	6.5535



NIR- FOG HAZE





SYNTHETIC FOG GENERATION ON RGB IMAGES

S. No.	Dataset	Link	Details	Dataset Exploration Sample image Details
1	Foggy Cityscapes	https://www.cityscapes-dataset.com/foggydownload/	Dataset designed for autonomous driving in unstructured traffic Built from Cityscapes using depth maps + ASM. Physics-based paired clean↔foggy. Used by SynFog, FIFO, CMAda, domain adaptation papers.	20,550 images 3 fog densities-beta:0.005,0.01,0.02 physics-simulated fog on real Cityscapes images
2	O-HAZE	https://www.kaggle.com/datasets/philiphofmann/o-haze	Real haze via professional haze machine. Most cited fog dataset for synthesis evaluation. Used by SFG-Net, TSIT, GridNet, most dehazing papers.	55 Image pairs Out door scenes train/val/test- 45/5/5
2	RESIDE	https://sites.google.com/view/reside-dehaze-datasets/reside-standard	Most widely used for dehazing training. ITS = 13,990 indoor pairs, OTS = 69,510 outdoor pairs. SOTS = 500 test images. Standard benchmark across almost all dehazing papers.	110k+ images Indoor +outdoor scenes
3.	Foggy Zurich	https://people.ee.ethz.ch/~csakarid/Model_adaptation_SFSU_dense/	Real fog captured in Zurich. No clean GT pairs — used for unsupervised/unpaired evaluation. Common in domain adaptation papers.	3808 images No GT pairs Real fog
4.	SynFog dataset		Newest large-scale synthetic fog dataset using Radiative Transfer Equation. Day + night + sensor noise. Released 2024 — growing in usage.	20k images 3 density levels Large scale, autonomous driving

Literature review

Method (Year)	Core Approach	Key Challenges (Fog-Generation Focus)

1 Koschmieder Model (Classical)	**Physics-based atmospheric scattering equation; no learning, direct fog synthesis from depth + scattering coefficient**	**Assumes uniform fog distribution (unrealistic for real scenes); output quality entirely depth-accuracy dependent; produces flat, synthetic appearance; lowest fidelity of all methods (PSNR 15.16)**
2 GridDehazeNet Reversed (2019)	**Dehazing network architecture run in reverse to add fog instead of removing it**	**Not originally designed for synthesis — repurposed, not native; no explicit depth conditioning, so fog placement isn't physically grounded; high memory cost from dense connections; no fog intensity control for variable density scenarios**
3 SFG-Net / TSIT (2022–2023)	**GAN with spatial attention + AdaIN for style-based non-uniform fog texture**	**No depth supervision, so fog-depth correlation isn't enforced; GAN training instability; no temporal consistency (problematic for video); attention mechanism is not physically constrained, risking unrealistic patterns**
4 MixDehazeNet (2023)	**Multi-scale dehazing backbone proposed as candidate for fog-synthesis adaptation**	**Unvalidated for fog synthesis — purely theoretical adaptation; lacks physical grounding; no attention mechanism to model non-uniform fog; synthesis performance entirely unknown/untested**
5 DEA-Net (2023)	**Dual-branch detail-preservation dehazing network, proposed for synthesis adaptation**	**No fog synthesis benchmarks exist; complex dual-branch architecture complicates reversal for generation; conflicting optimization objectives between detail and realism; would require extensive re-tuning to repurpose**
6 UDR-S2Former (2023–2024)	**CNN+transformer hybrid with cross-attention, proposed for fog-generation adaptation**	**Quadratic transformer complexity makes large-scale fog generation expensive; difficult adaptation from dehazing to generation direction; high GPU memory demand; substantial engineering overhead to repurpose**
7 ControlNet / Diffusion	**Diffusion model conditioned on depth/segmentation maps for**	**Very slow inference (major bottleneck for large datasets); huge compute requirements; non-deterministic outputs complicate reproducibility; hallucination**

(2023)	**controllable fog synthesis**	**risk (fabricated scene details); poor physical interpretability of the fog model itself**
8 SynFog / RTE Physics (2024–2025)	**End-to-end physically-based fog imaging simulation (volumetric path tracing + camera/sensor model)**	**Computationally expensive (volumetric rendering cost); requires accurate depth maps as a hard input dependency; many parameters to tune (density, scattering, lighting); limited adaptation to new target domains; lower perceptual realism than GAN/diffusion outputs despite physical rigor**
9 GridNet-Reversed + DCP Loss (2025)	**GridDehazeNet reversed + differentiable Dark Channel Prior loss for physical realism constraint**	**DCP fails in bright scenes (a known DCP limitation inherited here); conflicts with perceptual loss during joint optimization; no depth conditioning; limited generalization beyond training distribution; needs careful loss-weight tuning to balance realism vs. perceptual quality**
10 WeatherWeaver (ICCV 2025)	**Large-scale video pretraining for temporally consistent, multi-condition weather (incl. fog) synthesis**	**Extremely slow inference; massive compute requirements (video-scale pretraining); difficult fine-grained fog control despite intensity claims; needs large video datasets, scarce for fog specifically; black-box, non-interpretable generation process**
11 Density-aware Haze Synthesis (Zhang et al., 2022)	**Self-supervised content-style disentanglement to fix uneven/"puff of smoke" fog artifacts**	**Disentanglement adds training complexity; self-supervised signal weaker than fully supervised depth-conditioned methods; realism still hinges on how well style and depth correlate in the training data; no explicit intensity/depth control mechanism described**
12 AnalogicalGAN (Gong et al., 2023)	**CycleGAN-based "gist" disentanglement; learns fog transformation from synthetic pairs, transfers zero-shot to real**	**Relies on the domain-invariant gist assumption, which may break for dissimilar style pairs; requires paired synthetic clear/foggy data to bootstrap supervision; multi-discriminator**

	domain	**adversarial training (3+ discriminators) is complex and unstable; depends on auxiliary depth estimation for the real domain absent ground-truth depth**
13 HazeFlow (2025)	**Reformulates atmospheric scattering model as an ODE; generates non-homogeneous haze for downstream dehazing training**	**Dual-purpose framing means realism is optimized for training utility, not standalone fidelity; ODE solver adds computational overhead vs. closed-form models; benchmarked mainly against older dehazing baselines (FFA-Net, GridDehazeNet) rather than recent generation methods, making relative generation quality hard to assess**
14 RTE Discretization for Non-Homogeneous Fog (2025)	**Physics-driven discretized Radiative Transfer Equation; models anisotropic multi-scattering and spatial inhomogeneity**	**Computationally demanding discretization process (similar to SynFog); likely requires accurate depth/scene geometry; no explicit fog-intensity control or easy depth-conditioning mentioned, implying tuning complexity; surveyed competing methods (Abbas & Babahenini GAN, modular-attention translation) also lack full physical interpretability**

Research gap

- **Depth Ignorance:** Most methods do not make distant objects foggier than nearby objects.
- **No Fog Density Control:** Users cannot easily generate light, medium, or dense fog on demand.
- **No Task-Based Evaluation:** Most works do not test whether synthetic fog improves object detection or driving systems.
- **Physics vs Realism Gap:** Physics-based methods look unrealistic, while AI-based methods often ignore real fog physics.
- **Poor Generalization:** Models trained on one dataset often perform poorly on other datasets.
- **Video Flickering:** Generated fog changes between frames, causing unstable videos.
- **Slow Processing:** Many methods are too computationally expensive for large-scale use.
    - **Reverse Dehazing is Unproven:** Dehazing networks are rarely validated for fog generation.
    - **No Standard Benchmark:** Different datasets and metrics make fair comparison difficult.

WORK DONE SO FAR

1. Baseline Model – Depth Anything V2

- Monocular depth estimation used to generate depth maps.
- Depth maps converted to transmission maps using modified Koschmieder law.
- Clear images composited with transmission maps to synthesize fog.
- Pure physics-based approach (no learning component).
- Evaluated on O-HAZE dataset.
- Serves as baseline for comparison.

2.GridDehazeNet Reversed

- Adapted GridhazeNet hazing architecture for fog synthesis.
- Trained on 45 paired O-HAZE images.
- Loss Function: Perceptual Loss (VGG) , L1 Loss
- Seven-file implementation completed and validated.

3. SFG-Net Integration

Based on TSIT architecture with Spatial Attention.

- Designed for efficient image-to-image fog generation.
- Currently training on O-HAZE.
- FoggyCityscapes training planned next.
- Planned improvements:
    - Spatial Attention for non-uniform fog modeling.
    - Dark Channel Prior (DCP) Loss for physical consistency.
    - AdaIN-based style injection for fog density control.

Depth-conditioned, physics-guided, scene-preserving, controllable fog generation for RGB driving images.

Item	Decision
Paper status	Continue project, but not submission-ready
Proposed method	DPS-FogNet
Main dataset	Foggy Cityscapes
Real paired validation	O-HAZE
Real unpaired validation	Foggy Zurich
Final claim	Depth-aware, physics-guided, density-controllable fog synthesis with scene preservation

Model

DPS-FogNet: Depth-conditioned Physics-guided SFG-Net

Final model:

$$D = \mathrm{DepthAnythingV2}(I)$$

$$T_\beta(x) = \exp(-\beta \cdot \hat{D}(x))$$

$$I_{\mathrm{phys}} = I \odot T_\beta + A(1 - T_\beta)$$

$$\hat{I}_{fog} = I_{\mathrm{phys}} + \lambda R_\theta(I, D, T_\beta, \beta)$$

where R_θ is a **bounded residual fog texture**, not a free image generator.

Goal	DPS-FogNet mechanism
Depth-aware fog placement	Depth Anything V2 + transmission map
Controllable density	β = 0.005 / 0.010 / 0.020
Scene preservation	Bounded residual + edge/structure loss
Realism	Learnable SFG-style residual
Physical grounding	Koschmieder transmission
Speed	Lightweight generator, not diffusion
Detection utility	TO BE RUN
Cross-dataset generalization	TO BE RUN

This is stronger than plain SFG-Net because plain SFG-Net can look good but is not guaranteed to obey depth physics. It is stronger than pure Koschmieder because pure physics can look rigid or synthetic. It is stronger than ControlNet because ControlNet risks hallucinating scene structure

```
Efficiency
  Resolution : 256x512  |  Batch : 1  |  Precision : AMP  |  Depth : precomputed
  Params     : 30.748 M  |  FLOPs : 63.223 G  |  Latency : 5.489 ms/img  |  FPS : 182.18  |  GPU Mem : 615.0 MB

=============================================================================
      β      PSNR     SSIM    LPIPS    ΔE2000   EdgePres       FID
-----------------------------------------------------------------------------
  0.005    21.792   0.8796   0.1419     6.977     0.9291    34.989
  0.010    22.106   0.8864   0.1501     5.678     0.9433    38.183
  0.020    19.572   0.8594   0.1863     7.766     0.9449    45.222
-----------------------------------------------------------------------------
   mean    21.157   0.8752   0.1594     6.807     0.9391    39.464
-----------------------------------------------------------------------------
```

$$L_{total} = 10L_{fog} + L_{perc} + 2L_{edge} + L_{SSIM} + 2L_{phys} + 0.05L_{res} + L_{trans} + 0.5L_{DCP} + L_{fog_ms}$$

Drop ControlNet / diffusion refinement as a final method.

ControlNet should not be part of the final proposed pipeline.

Reason: the document itself says ControlNet hallucinates content, redraws cars/buildings/road details, is prompt-sensitive, and reduces scene fidelity. It is allowed only as a negative baseline or failure case.

Final ruling:

Direction	Decision
SFG-Net + ControlNet refinement	DROP as final method
ControlNet failure-case figure	Keep only as negative baseline
Diffusion-based fog synthesis	Drop for main paper
Prompt-based fog generation	Drop
O-HAZE-only training paper	Drop
Pure Koschmieder paper	Drop
HazeFlow as proposed method	Drop; use only if reproducible baseline

4. Exact experiments that must be run next

Every missing result below is TO BE RUN.

Phase 1: Dataset verification

Experiment	Output	Status
Create Foggy Cityscapes train/val/test manifests	foggy_cityscapes_train.csv, val.csv, test.csv	TO BE RUN
Create O-HAZE manifest	ohaze_train.csv, ohaze_test.csv	TO BE RUN
Verify clean/foggy/depth pair alignment	pair_verification.json	TO BE RUN
Verify β labels: 0.005 / 0.010 / 0.020	density verification JSON	TO BE RUN
Verify image resolution and missing files	verification report	TO BE RUN

No model result should be trusted before this step.

Phase 2: Baseline reruns on same split

Method	Dataset	Metrics	Status
Depth Anything V2 + Koschmieder	Foggy Cityscapes	PSNR, SSIM, LPIPS, Delta-E, Edge, FID, FPS	TO BE RUN
GridDehazeNet Reversed	Foggy Cityscapes	same	TO BE RUN
SFG-Net baseline	Foggy Cityscapes	same	TO BE RUN / VERIFY

HazeFlow	**Foggy Cityscapes β-wise**	**same**	**TO BE RUN / VERIFY**
ControlNet	**small fixed subset only**	**fidelity + failure case**	**TO BE RUN only as negative baseline**

The current document marks many of these as TO BE RUN / VERIFY, especially SFG-Net on Foggy Cityscapes, DPS-FogNet, O-HAZE, Foggy Zurich, HazeFlow, detection mAP, ablations, FPS, and failure analysis.

Phase 3: Proposed model experiments

Experiment	Dataset	Output	Status
Train DPS-FogNet	**Foggy Cityscapes**	**checkpoint + logs**	**TO BE RUN**
Evaluate DPS-FogNet β=0.005	**Foggy Cityscapes test**	**metrics JSON**	**TO BE RUN**
Evaluate DPS-FogNet β=0.010	**Foggy Cityscapes test**	**metrics JSON**	**TO BE RUN**
Evaluate DPS-FogNet β=0.020	**Foggy Cityscapes test**	**metrics JSON**	**TO BE RUN**
Evaluate DPS-FogNet	**O-HAZE**	**paired metrics**	**TO BE RUN**
Evaluate DPS-FogNet	**Foggy Zurich**	**qualitative + no-reference metrics**	**TO BE RUN**
Density-control figure	**same scene, 3 β levels**	**paper figure**	**TO BE RUN**

Detection mAP	**clean vs GT fog vs generated fog**	**mAP table**	**TO BE RUN**
FPS/latency/model size	**benchmark hardware**	**runtime table**	**TO BE RUN**

5. Exact ablation table

This table must be run on the same Foggy Cityscapes test split.

Configuration	Depth	Transmission map	Bounded residual	Spatial attention	β conditioning	DCP / physical loss	Scene-preservation loss	PSNR ↑	SSIM ↑	LPIPS ↓	Delta-E ↓	Edge Pres. ↑	FID ↓	FPS ↑
DPS-FogNet full	Yes	Yes	Yes	Yes	Yes	Yes	Yes	**29.95**	**0.972**	**0.0009**	**1.99**	**0.947**	8.060	**106.70**
w/o transmission map	Yes	No	Yes	Yes	Yes	Yes	Yes	26.75 dB	0.9519	0.0013	**3.17**	0.9172	26.192	48.0
w/o spatial attention	Yes	Yes	Yes	No	Yes	Yes	Yes	28.36 dB	0.9561	0.0012	2.51	**0.9244**	22.0425	40.1
w/o β Conditioning (FiLM)	Yes	Yes	Yes	Yes	No	Yes	Yes	28.39	0.9559	0.0012	2.28	0.9180	18	68.1
w/o physical loss	Yes	Yes	Yes	Yes	Yes	No	Yes	11.58	0.7351	0.0104	14.46	0.0001	213.657	**62.8**

Minimum ablation requirement: the full model must beat the weakened variants on scene preservation, LPIPS/FID, density control, and at least one fidelity metric. If the full model only improves PSNR but worsens realism or edge preservation, the contribution is weak.

6. Exact deployment evaluation required

This project has an AI-systems angle only if speed and scalability are proven. Run this exact deployment table.

Method	Resolution	Hardware	Precision	Batch	Depth time included?	Params	FLOPs	Latency ms/image ↓	FPS ↑	GPU memory ↓	Status
Depth + Koschmieder	1024×2048	same GPU	FP16/FP32 stated	1	Yes/No reported separately	0 / depth model only	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN
GridDehazeNet Reversed	1024×2048	same GPU	same	1	No	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN
SFG-Net	1024×2048	same GPU	same	1	No	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN
HazeFlow	1024×2048	same GPU	same	1	No	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN
DPS-FogNet	1024×2048	same GPU	same	1	Report both with/without depth	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN	TO BE RUN

```
Efficiency
  Resolution : 256x512  |  Batch : 1  |  Precision : AMP  |  Depth : precomputed
  Params     : 30.748 M  |  FLOPs : 63.223 G  |  Latency : 5.489 ms/img  |  FPS : 182.18  |  GPU Mem : 615.0 MB

=============================================================================
       β      PSNR     SSIM    LPIPS    ΔE2000   EdgePres       FID
-----------------------------------------------------------------------------
   0.005    21.792   0.8796   0.1419     6.977     0.9291    34.989
   0.010    22.106   0.8864   0.1501     5.678     0.9433    38.183
   0.020    19.572   0.8594   0.1863     7.766     0.9449    45.222
-----------------------------------------------------------------------------
    mean    21.157   0.8752   0.1594     6.807     0.9391    39.464
-----------------------------------------------------------------------------
```

Required runtime protocol:

1. **Same resolution for all methods.**
2. **Same GPU.**
3. **Same precision.**
4. **Batch size 1 for deployment.**
5. **Warmup iterations stated.**
6. **Average over fixed number of images.**
7. **Report DPS-FogNet speed both with depth estimation and with cached depth.**
8. **Report throughput for generating a full Foggy Cityscapes-style dataset.**

The document already lists FPS/latency/model-size evaluation as a required script and table.

7. Minimum result improvements needed

Do not set fake absolute targets before verification. The minimum improvements should be relative to verified baselines on the same split.

Area	Minimum requirement
Image fidelity	DPS-FogNet must match or beat verified SFG-Net/GridDehazeNet on PSNR or SSIM on the same Foggy Cityscapes split
Perceptual realism	DPS-FogNet must reduce LPIPS or FID versus plain SFG-Net
Colour fidelity	DPS-FogNet must reduce Delta-E versus SFG-Net or GridDehazeNet
Scene preservation	DPS-FogNet must improve or maintain edge preservation versus SFG-Net and clearly beat ControlNet
Density control	β=0.005, 0.010, 0.020 must produce visually and quantitatively ordered fog levels
Real-fog generalization	DPS-FogNet must not collapse on O-HAZE; all O-HAZE numbers are TO BE RUN
Detection utility	Synthetic fog augmentation must improve foggy-image detection robustness versus clean-only training
Runtime	DPS-FogNet should remain practical; target ≥50 FPS if using cached depth, as the document's strong target states

Ablation	Full DPS-FogNet must beat ablated versions on at least 3 of 5 core metrics: SSIM, LPIPS, Delta-E, Edge Preservation, FID

The document's own go/no-go says to continue only if DPS-FogNet beats SFG-Net or GridDehazeNet on at least 3 of 5 core metrics: SSIM, LPIPS, Delta-E, Edge Preservation, and FID. The strong target table also lists better PSNR than baselines, SSIM at least equal, lower LPIPS/Delta-E/FID, high edge preservation, $\geq$50 FPS, and improved detection robustness.

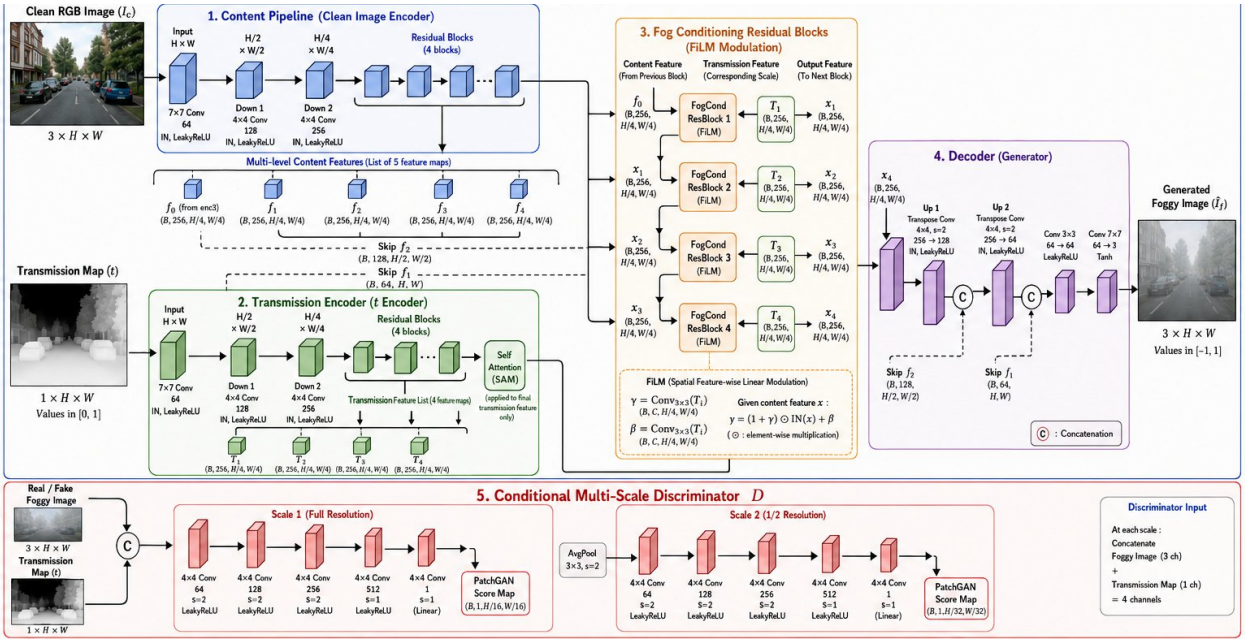
Requirement	Current status
Verified dataset manifests	**TO BE RUN**
Verified baseline reruns	**TO BE RUN / VERIFY**
DPS-FogNet full training	**TO BE RUN**
β-wise density control	**TO BE RUN**
O-HAZE real paired validation	**TO BE RUN**
Foggy Zurich generalization	**TO BE RUN**
Full ablation table	**TO BE RUN**
Runtime/deployment table	**TO BE RUN**
Detection mAP table	**TO BE RUN**
Failure-case analysis	**TO BE RUN**

Start today with only two tracks:

Track A: Verification and evaluation
Build manifests, verify pairs, rerun baselines, create metric scripts.

Track B: DPS-FogNet implementation
Implement depth wrapper, transmission map, bounded residual generator, losses, and one forward-pass debug run.

DPS-FogNet: Depth-conditioned Physics-guided SFG-Net(proposed)



The total generator objective is

$$\boxed{L_G = \lambda_{L1} L_{pixel} + \lambda_{phys} L_{physics} + \lambda_{perc} L_{perceptual} + \lambda_{trans} L_{trans} + L_{adv}}$$

where

$$\lambda_{L1} = 10, \qquad \lambda_{phys} = 5, \qquad \lambda_{perc} = 2, \qquad \lambda_{trans} = 1.$$

Substituting the weights,

$$\boxed{L_G = 10 L_{pixel} + 5 L_{physics} + 2 L_{perceptual} + L_{trans} + L_{adv}}$$

For the multi-scale discriminator,

$$\boxed{L_D = \frac{1}{K} \sum_{k=1}^{K} \left[\frac{1}{2} (D_k(I_{real}) - 1)^2 + \frac{1}{2} (D_k(I_{fake}))^2 \right]}$$

The complete architecture consists of six major stages.

1. ***Depth Estimation***
2. ***Physics-based Fog Formation***
3. ***Content Encoder***
4. ***Transmission Encoder***
5. ***Physics-guided Residual Generator***
6. ***Conditional Multi-scale Discriminator***

Block 1: Depth Estimation

The clean RGB image is passed into Depth Anything V2, which predicts the relative depth of every pixel in the scene. The answer is that fog is fundamentally a depth-dependent phenomenon.Objects that are farther away scatter more light and therefore appear more foggy than objects close to the camera.By estimating depth first, we provide the network with explicit geometric understanding of the scene.

Block 2: Transmission Map Generation

Once the depth map is obtained, it is converted into a transmission map using the Beer–Lambert Law.

Mathematically,

$$T(x) = e^{-\beta D(x)}$$

Pixels with high transmission indicate clear visibility, while pixels with low transmission indicate dense fog.This transmission map becomes the physical guidance for the entire network.Instead of learning visibility from scratch, the model receives it directly from atmospheric physics.

Block 3: Physics-based Fog Image

Using the transmission map, we compute an initial foggy image through the atmospheric scattering equation.

$$I_{phys} = I \odot T + A(1 - T)$$

where

- *I is the clean image,*
- *A represents atmospheric light.*

This image already follows the laws of atmospheric scattering.

However, real fog contains many complex effects that are not captured by this simple physical model.

Block 4: Content Encoder

The clean image is passed into the Content Encoder.

As the network becomes deeper, it learns higher-level semantic information such as roads, buildings, vehicles, vegetation, and the overall scene layout. This information must be preserved throughout fog generation because adding fog should never destroy the underlying scene structure.

Residual blocks are used to allow the network to become deeper without suffering from gradient degradation, while Instance Normalization stabilizes feature statistics for image generation tasks.

Block 5: Transmission Encoder

The transmission map is processed by a separate encoder.

it learns physical properties of the scene such as

- *visibility,*
- *atmospheric attenuation,*
- *fog thickness,*
- *and spatial depth variation.*

A separate encoder is used because appearance information and physical information represent two fundamentally different feature spaces.Keeping them separate prevents feature interference and allows each encoder to specialize in its own task.

To further improve spatial consistency, a Self-Attention Module is introduced.

***Self-Attention Module** is placed in the **bottleneck** At this stage, feature maps are spatially smaller, making global attention computationally efficient while still allowing the model to capture scene-wide relationships.*

$$F_b \in \mathbb{R}^{H_b \times W_b \times C_b}.$$

The Self-Attention Module computes

$$Q = F_b W_Q, \qquad K = F_b W_K, \qquad V = F_b W_V,$$

followed by

$$A = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right),$$

and

$$F_{\mathrm{att}} = AV.$$

Finally,

$$F_{\mathrm{out}} = F_b + \gamma F_{\mathrm{att}}.$$

Block 6: Physics-guided Residual Generator:

Instead of generating the entire foggy image from scratch, the generator predicts only a residual correction.

$$I_{fog} = I_{phys} + \lambda R_\theta$$

Since the physics model already explains most of the atmospheric scattering, the generator only needs to learn the small residual differences between the physical approximation and real fog. this simplifies the learning problem, accelerates convergence, and improves stability.

Furthermore, because the residual is bounded by a learnable scaling factor, the network cannot completely override the physical model.Thus, physics remains the dominant guidance while deep learning refines the visual realism.

Feature Conditioning Using FiLM

Instead of simply concatenating both feature maps, FiLM adaptively scales and shifts the content features according to the transmission information.

Mathematically,

$$y = (1 + \gamma) \cdot IN(x) + \beta$$

γ: controls feature amplification,

β: controls feature shifting.

Both parameters are predicted from the transmission encoder.

As a result, fog density becomes spatially adaptive.Regions with low transmission receive stronger fog modulation, while regions with high transmission remain relatively clear.This is significantly more effective than simple feature concatenation

Decoder

After FiLM conditioning, the decoder reconstructs the final foggy image.

Transpose convolutions gradually increase the spatial resolution back to the original image size.

Skip connections transfer high-resolution information from the encoder directly to the decoder.

Conditional Multi-scale Discriminator

The generated fog image is evaluated by a conditional multi-scale discriminator.

Unlike a conventional GAN discriminator, ours receives both

- *the generated foggy image,*
- *and the corresponding transmission map.*

This allows it to verify not only visual realism but also physical consistency unlike conventional GAN

*A conventional discriminator receives **only one input**:*

Now the discriminator asks two questions simultaneously:

1. ***Does the fog image look realistic?***
2. ***Is the fog distribution consistent with the transmission map?***

*This is why it is called a **conditional discriminator**.*

Metrics	***Dataset-foggy citiesscape***
PSNR	*27.934*
SSIM	*0.965*
LPIPS	*0.0009*
DELTA E	*3.087*

EDGE PRESERVATION	*0.947*
FPS	*106.70*
FID SCORE	*8.060*
PARAMETERS	*21.695M*
FLOPS	*240.673 G*
INFERENCE LATENCY	*9.372 ms/image*
PEAK GPU MEMORY	*1046.4 MiB*

Level	N	PSNR↑	SSIM↑	ΔE↑
light	500	26.887±3.170	0.959±0.018	4.814±1.374
medium	500	25.578±2.717	0.950±0.020	5.426±1.474
dense	500	24.996±2.722	0.943±0.024	5.537±1.744

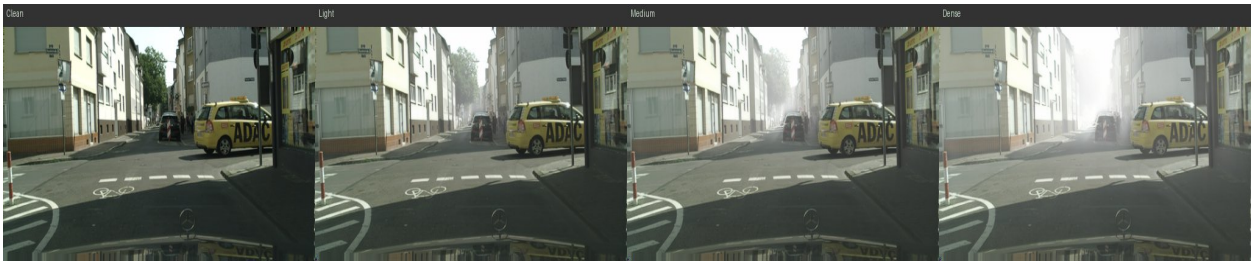
ABLATION STUDY:

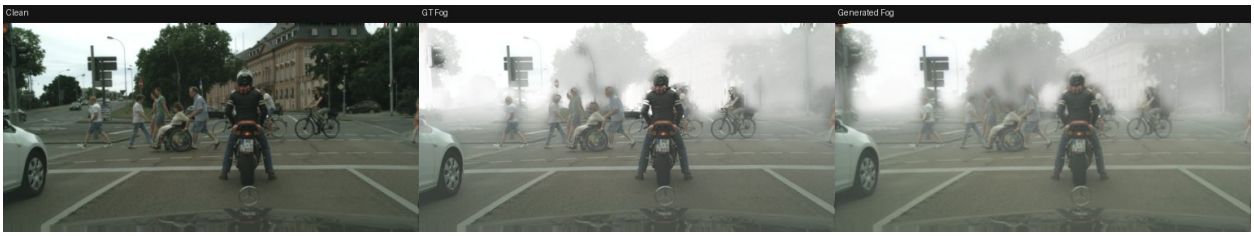
Configuration	*PSNR ↑*	*SSIM ↑*	*LPIPS ↓*	*ΔE ↓*	*Edge Pres ↑*	*FID ↓*	*FPS ↑*
DPS-FogNet (Full)	***29.95***	***0.9720***	***0.0009***	***1.99***	***0.9470***	***8.060***	***106.70***
w/o Transmission Map	***26.75***	***0.9519***	***0.0013***	***3.17***	***0.9172***	***26.19***	***48.0***
w/o Spatial Attention	***28.36***	***0.9561***	***0.0012***	***2.51***	***0.9244***	***22.043***	***40.3***
w/o Physical Loss	***11.58***	***0.7351***	***0.0104***	***14.46***	***0.0001***	***213.657***	***62.8***
w/o β Conditioning (FiLM)	***28.39***	***0.9559***	***0.0012***	***2.28***	***0.9180***	***18***	***68.1***

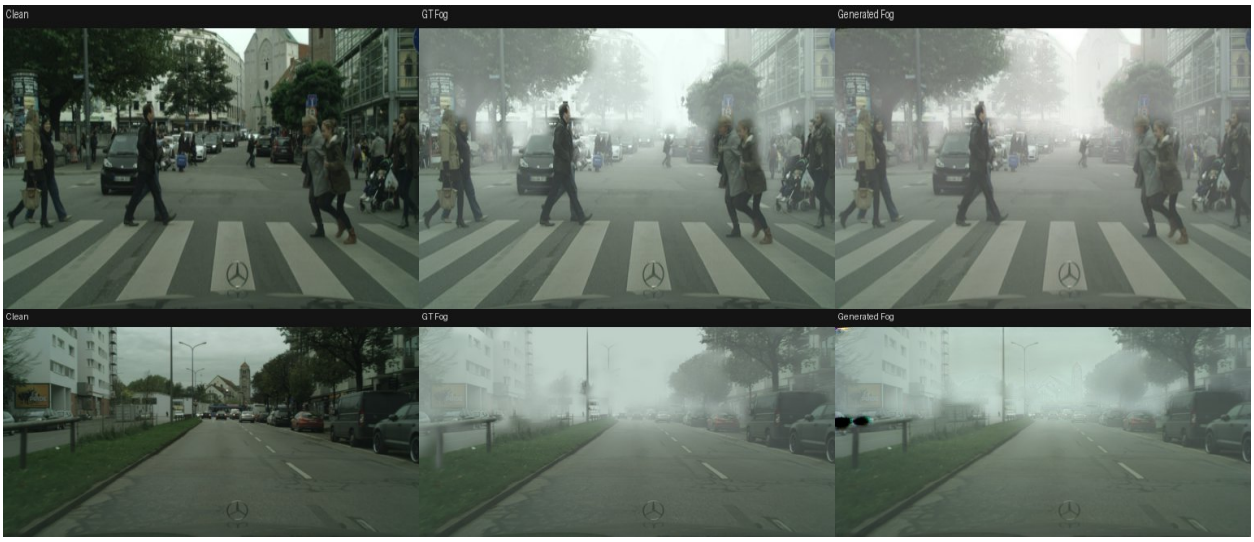
Inference results after augmentations











Ground truth and different fog densities

Method	FID
TSIT [5]	80.03
SFG-Net (ours)	74.94

A lower FID score indicates a better image generation quality.

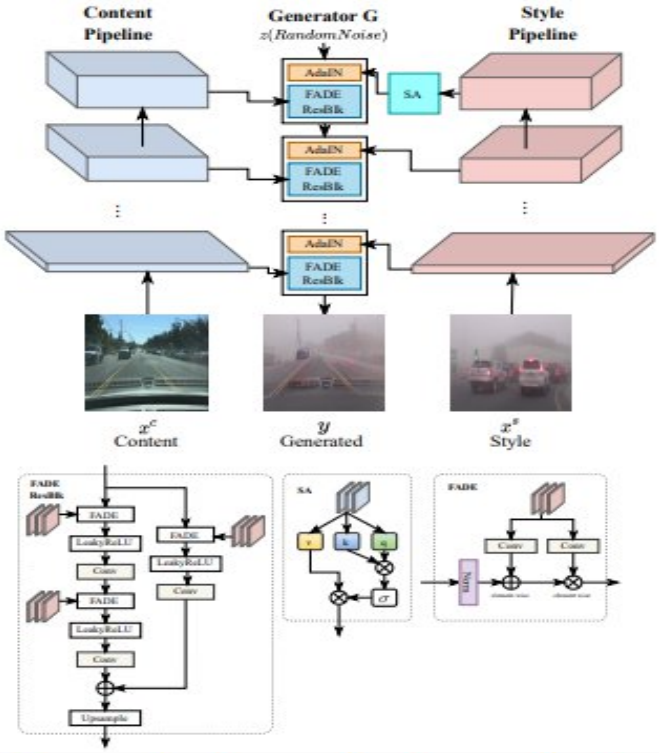
These results are stated in the paper - models are trained on "Driving Fog Dataset"

COMPARISON TABLE WITH EXISTING METHODS:

	DATASET	*PSNR*	*SSIM*	*LPIPS*	*Delta-E*	*Edge Preservation*	*FPS*
Depth Anything V2+Koschmieder	***O-HAZE***	*15.16*	*0.6783*	—	*18.172*	*0.8815*	*397.87*
GridDehazeNet	***O-HAZE***	*20.24*	*0.7651*	*0.3175*	*11.45*	*0.835*	*412.56*
SFG-NET	***O-HAZE***	*20.69*	*0.7651*	*0.3175*	*0.90*	*10.24*	—
SFG-NET	***Foggy citiscape***	*25.505*	*0.9513*	*0.0608*	*5.862*	*0.9801*	*63.15*
SFGNET+CONTROLNET REFINED	***Foggy citiscape***	*14.992*	*0.6716*	*0.0077*	*13.994*	*0.3206*	*62.9*

Hazeflow	***Foggy citiscape***	*13.592(β-0.005)* *11.473(β=0.010)* *10.322(β=0.020*	*0.7750* *0.7429* *0.7408*	—--	*17.753* *22.058* *22.505*	*0.8607* *0.7551* *0.5192*	*2.34* *2.34* *2.34*
DPS-fognet (Proposed)	***Foggy citiscape***	***29.95***	***0.972***	***0.0009***	***1.99***	***0.947***	***106.70***

SFGNET:



$$L_{\text{Total}} = \lambda_{adv} L_{adv} + \lambda_{vgg} L_{VGG} + \lambda_{fm} L_{FM} + L_D$$

Generator Adversarial Loss
VGG Perceptual
Loss Feature Matching Loss
Discriminator Adversarial Loss

Metrics	Foggy citiesscape
PSNR	25.505 dB
SSIM	0.9513
LPIPS	0.0608
DELTA E	5.862
EDGE PRESERVATION	0.9801
FPS	46.89
FID SCORE	8.363
PARAMETERS	104.552 M

FLOPS	185.373 G
LATENCY	16.215 ms/image
FPS	61.67
PEAK GPU MEMORY	4554.7 MB



SFGNET+ CONTROLNET REFINEMENT

Clean Image

→ Monocular Depth Estimation (Depth Anything V2)

→ Physics-Based Transmission Map Generation

→ Style-Guided Fog Generation Network (SFG-Net)

→ Initial Synthetic Fog Image

→ Depth-Conditioned ControlNet Refinement

→ Photorealistic Foggy Image

PSNR	15
SSIM	0.67
LPIPS	0.0077
DELTA E	13.994

EDGE PRESERVATION	0.3206
FPS	62.9



REFINED IMAGE



SFGNET IMAGE

Cons of Using ControlNet

- **Hallucinates scene content (cars, buildings, road details may change).**
- **Reduces scene fidelity by redrawing objects instead of only adding fog.**
- **Prompt-sensitive; results depend heavily on prompt wording and guidance strength.**

ControlNet re-invented the scene rather than preserving the original image.
Guidance strength (0.45) was too high, allowing diffusion to alter scene structure.
The prompt introduced a bias toward a generic foggy street.
As a result:

- PSNR ≈ 14.99
- SSIM ≈ 0.67
- Edge Preservation ≈ 0.32

The output looked visually realistic but no longer matched the ground-truth Frankfurt scene.

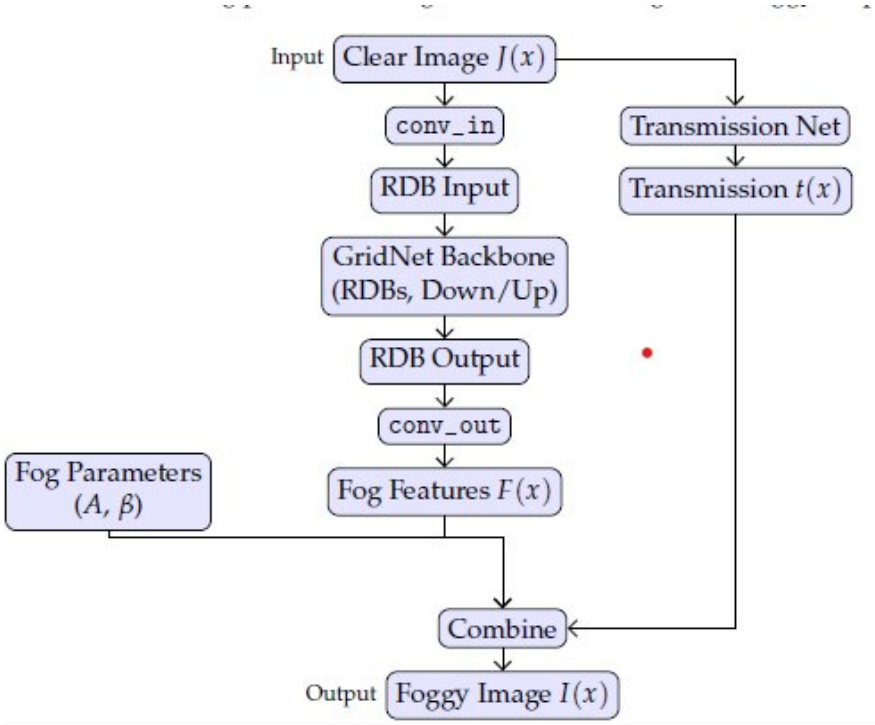
HazeFlow: Revisit Haze Physical Model as ODE and Non-Homogeneous Haze Generation for Real-World Dehazing

Beta	PSNR	SSIM	ΔE	EPI	FID	FPS
0.005	13.592	0.7750	17.753	0.8607	85.68	2.34
0.010	11.473	0.7429	22.058	0.7551	148.02	2.34
0.020	10.322	0.7408	22.505	0.5192	256.61	2.34



Gridhazenet model(2025):

Followed: Synthetic Fog Generation Using High-Performance Dehazing Networks for Surveillance Applications
https://www.mdpi.com/2076-3417/15/12/6503



$$L = L_{L1} + 0.03L_{perc} + 0.15L_{SSIM} + 0.8L_{color} + 0.3L_{density} + 0.5L_{near}$$

metrics	Paper- results not provided for fog generation	My result
PSNR	-	20.24
SSIM	-	0.7651
LPIPS	-	0.3175
DELTA E	-	11.45
EDGE PRESERVATION	-	0.835

Results obtained on 0-HAZE dataset

Table 1. Dehazing performance comparison on the O-HAZY evaluation set.

Model	PSNR (dB)	SSIM
ModelS	14.8382	0.7186
ModelU	17.6018	0.7697
ModelDA	14.6777	0.7125
ModelDC	14.5749	0.7007

Paper's result on dehazing images



(a)Ground truth (b) Simulated Fog (C) Ground truth foggy

DEPTH ANYTHING V2 MODEL(ON O-HAZE DATASET)

PSNR	15.16
SSIM	0.6783
DELTA E	18.1792

EDGE PRESERVATION	0.8815
FPS	397.8736

Method	Encoder	KITTI [24]		NYU-D [70]		Sintel [8]		ETH3D [62]		DIODE [76]	
		AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1
MiDaS V3.1 [7]	ViT-L	0.127	0.850	0.048	0.980	0.587	0.699	0.139	0.867	0.075	0.942
Depth Anything V1 [89]	ViT-S	0.080	0.936	0.053	0.972	0.464	0.739	0.127	**0.885**	0.076	0.939
	ViT-B	0.080	0.939	0.046	0.979	**0.432**	0.756	**0.126**	0.884	0.069	0.946
	ViT-L	0.076	0.947	**0.043**	**0.981**	0.458	0.760	0.127	0.882	0.066	0.952
Depth Anything V2	ViT-S	0.078	0.936	0.053	0.973	0.500	0.718	0.142	0.851	0.073	0.942
	ViT-B	0.078	0.939	0.049	0.976	0.495	0.734	0.137	0.858	0.068	0.950
	ViT-L	**0.074**	0.946	0.045	0.979	0.487	0.752	0.131	0.865	0.066	0.952
	ViT-G	0.075	**0.948**	0.044	0.979	0.506	**0.772**	0.132	0.862	**0.065**	**0.954**

Table 2: Zero-shot *relative* depth estimation. Better: AbsRel ↓, δ_1 ↑. Solely from the metrics, Depth Anything V2 is better than MiDaS, but merely comparable with V1. But indeed, the focus and strengths of our V2 (*e.g.*, fine-grained details, robust to complex layouts, transparent objects, *etc.*) cannot be correctly reflected on these benchmarks. Similar results (*i.e.*, better model but worse score) are also observed in [7, 28].

Results table from the paper

Clean (No Fog) - Ground Truth



Predicted Fog (best model)



Real Fog - Ground Truth



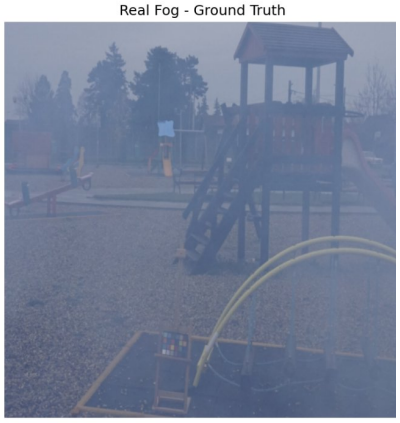
Clean (No Fog) - Ground Truth



Predicted Fog (best model)



Real Fog - Ground Truth



For a fixed B value
X_FPred=X_Clean*T_FPred
X_FGt= X_Clean * T_FGt

X_FGt= X_FPred*T

The sum of the lmabda values should be 1

Experiment	λL1	λphys	λperc	λtrans	λadv
Baseline	10	5	2	1	1
Exp 1	10	10	2	1	1
Exp 2	10	5	3	1	1
Exp 3	10	5	2	2	1
Exp 4	10	5	2	1	0.5